

Tai Chi Balance Analysis: Novice vs. Trained

Code and outputs: https://github.com/nvanvlasselaer/Thai_chi (All rights reserved)

1. Selected Balance-Relevant Events:

1. Trunk Rotations / Weight Shifts:

Description: Major rotational transitions characterized by large trunk (pelvis-chest) relative yaw (z-axis) excursion accompanied by a peak in chest angular velocity.

Balance Relevance: These movements require the practitioner to transfer their weight laterally while managing rotational momentum. This challenges the medial-lateral stability limits and requires dynamic, coordinated decoupling of the upper and lower body to prevent falling.

2. Monopodal Stance with Unilateral Knee Flexion:

Description: Windows where knee flexion exceeds 60 degrees. As observed in the sequence, this occurs during a monopodal stance where the subject performs a unilateral hip flexion together with a knee flexion (first left, then right). Later in the sequence, this includes a variation where the knee extends while keeping the hip in flexion.

Balance Relevance: Lifting one leg into hip and knee flexion reduces the base of support to a single foot. This forces the CoM to shift over the stance leg, creating a challenge to both medial-lateral (ML) stability and antero-posterior (AP) control. The ability to stabilize the trunk and pelvis during these unilateral, single-leg phases can show postural competence.

2. Alignment of Novice and Trained Recordings

Tai Chi movements are fluid and self-paced, a novice and an expert will naturally perform the sequence at different speeds. To compare them, I used **Dynamic Time Warping (DTW)** from Madwick sensor fused kinematics (possibly even removing the need to perform the sequence while watch the instruction video)

1. I extracted discrete trunk rotation events from the Novice recording
2. Next for each novice event, generated a multi-dimensional signature matrix comprising of a set of detrended trunk-pelvis yaw, lumbar roll, hand angular velocities, and bilateral knee flexion angles.
3. Then slid this template across the Trained recording and computed the DTW distance. DTW non-linearly stretches and compresses the time axis to find the best morphologic match, allowing us to find the exact corresponding trained movement module regardless of speed execution differences.

3. Balance Metric Outputs and Interpretation

A. Medio-Lateral (ML) and Antero-Posterior (AP) Sway:

Estimates micro movements of the CoM by calculating the variance of horizontal linear acceleration at the lumbar IMU during the quasi-static stabilization phase immediately following an event.

Result: Although small, during trunk rotations, the Trained practitioner exhibited lower AP sway (mean $0.0011g^2$ vs Novice $0.0027g^2$) but higher ML sway (mean $0.00078g^2$ vs Novice $0.00026g^2$).

Interpretation: Lower AP sway in the expert indicates superior sagittal plane stability and grounding. Interestingly, the expert's higher ML sway variance should not be interpreted as instability. Instead, it could reflect functional flexibility (the expert allows fluid, continuous lateral weight transfer, whereas the novice demonstrates less freedom). The novice's rigidly low ML variance could reflect stiffness, and an over-control strategy utilized when a practitioner is unsure of their balance limits. These differences are small and need to be verified in a larger sample.

B. Trunk-Pelvis Coordination Lag

The timing difference (cross-correlation lag) between the pelvic and trunk rotational trajectories (yaw).

Result: The Trained practitioner showed shorter, more tightly coupled lag times (mean -0.18s) compared to the Novice (mean -0.42s).

Interpretation: This is a measure of kinetic chain efficiency. The expert coordinates their trunk and pelvis almost simultaneously, shifting them as a unified but articulated column. The novice exhibits longer delays, pointing to a disjointed, uncoupled postural control strategy where segments must wait for one another to re-establish balance.

C. Left-Right Asymmetry

The absolute difference between the left and right legs across all detected knee flexion events (e.g., difference in peak flexion degrees or stabilization time).

Result: The Novice showed a peak knee flexion asymmetry of 18.1° , whereas the Trained practitioner was more symmetric with an 11.2° difference.

Interpretation: Relevant for bilateral single-leg balance confidence. The trained practitioner executes the unilateral hip/knee flexion with more equal magnitude and stability regardless of which leg is lifted.

D. Movement Smoothness (Dimensionless Angular Jerk)

Jerk is the rate of change of acceleration. Normalized dimensionless jerk quantifies the overall smoothness and fluidity of the trunk's weight shift, independent of the movement's duration or amplitude.

Result: Both practitioners showed relatively similar dimensionless jerk magnitudes (Log10 9.1) during the trunk rotations, though the novice had slightly higher baseline orientation variability.

Interpretation: While jerk is an excellent measure of overall movement fluidity, the similarity here suggests that the novice is successfully replicating the macro-kinematics (gross smoothness) of the slow Tai Chi movements.

4. Dataset Observations

Limitation of Absolute Yaw Metrics: The provided dataset relies exclusively on ACC and GYR data, lacking MAG inputs. Without a MAG to reference Earth's magnetic north, the initial yaw axis (rotation around the gravity vector) remains completely unobservable and is prone to integration drift over time.

To mitigate drift in this analysis, I applied a 0.05 Hz high-pass filter to the yaw signals. This cutoff was chosen to remove the slow-accumulating drift while trying to stay below the typical frequencies of Tai Chi movements (>0.5 Hz), preserving the kinematic signal.

This drift did not affect AP and ML joint angles much (such as knee flexion, hip flexion, or trunk-pelvis roll and pitch) as the pitch and roll orientations are continuously stabilized by the gravity vector measured by the ACC.

Sensor Alignment Strategy: Due to the absence of a MAG and the lack of a clear, predefined calibration pose (such as an N-pose or T-pose) in the recordings, I used the somewhat neutral posture at the beginning of the Tai Chi sequence to roughly align the IMU sensors to the body segments.

For future registrations, when recording with 6-DOF IMUs, determining the absolute (initial) heading is important. To solve this in future data collections, two strategies can be used:

Anatomical Alignment: By physically placing and aligning the sensors in the exact orientation of the underlying body segments. But while straightforward, this method is highly prone to human placement errors.

Functional Calibration Movement: Having the subject perform a predefined set of 1D calibration movements (e.g., pure hip flexion, pure trunk flexion) at the start of the recording. This allows for more accurate alignment the IMU coordinates to the anatomical body segments using kinematic constraints.

5. Workflow Scalability

By implementing automatic detection and alignment strategies (such as *detect_trunk_rotation_events*, *find_knee_flexion_windows*, *signature_matrix*, and *find_trained_match*), practically all movements of interest can be automatically extracted from continuous, multi-IMU recordings. These automatically detected events can be visually verified by either recreating a 3D stick-figure using forward kinematics (`animate_kinematics.py`) or by directly inspecting the joint kinematics plots (`plot_kinematics.py`). Making this balance analysis workflow robust and highly scalable to larger datasets.

6. Future Work: 2-Month Research Stay Proposal

1. **Implementation of Functional Calibration:** developing and validating the "Functional Calibration Movement" strategy mentioned above. This involves designing a brief, standardized movement protocol (e.g., isolated hip flexion, isolated trunk flexion) and writing the optimization code to map the IMU sensor frames directly to the anatomical joint axes.
2. **Scaling & Statistics:** Going from a two-subject proof of concept to a larger cohort. Because the event detection and DTW alignment are fully automated, bulk processing is feasible. The goal would be to establish statistically significant normative bands for the developed metrics.
3. **Single-Camera 3D Kinematics for Remote Assessment:** Since David mentioned a clinical goal is to analyse Tai Chi in a remote / at-home setting, exploring the reliability of markerless single-camera 3D kinematic reconstruction can be more practical. I have already successfully reconstructed initial kinematics from the provided video using GVHMR (see video in repo). However, its accuracy must be validated against optical tracking and compared to IMU sensors for choosing future focus.
(what has the best ease of use vs accuracy balance: 2D video or IMU?)
Optionally; using the optical tracking data as ground truth, the GVHMR machine learning model can be fine-tuned to ensure higher fidelity remote capture.
4. **Non-Linear Dynamics and Local Dynamic Stability (Later Stages):** As the dataset scales to include a large number of repetitive Tai Chi movement cycles, the analysis can evolve beyond, linear metrics to explore non-linear dynamics. Specifically, we can explore the use of for example Phase Space Reconstruction and the Rosenstein algorithm to calculate the largest Lyapunov exponent (LyE) from the continuous IMU data. This has the potential to quantify a practitioner's local dynamic stability, measuring how well they maintain consistent, controlled movement trajectories despite continuous internal and external perturbations. We can try to evaluate this stability on a univariate (per-joint) basis to pinpoint specific anatomical variation or use a multivariate embedding to assess whole-body coordination.

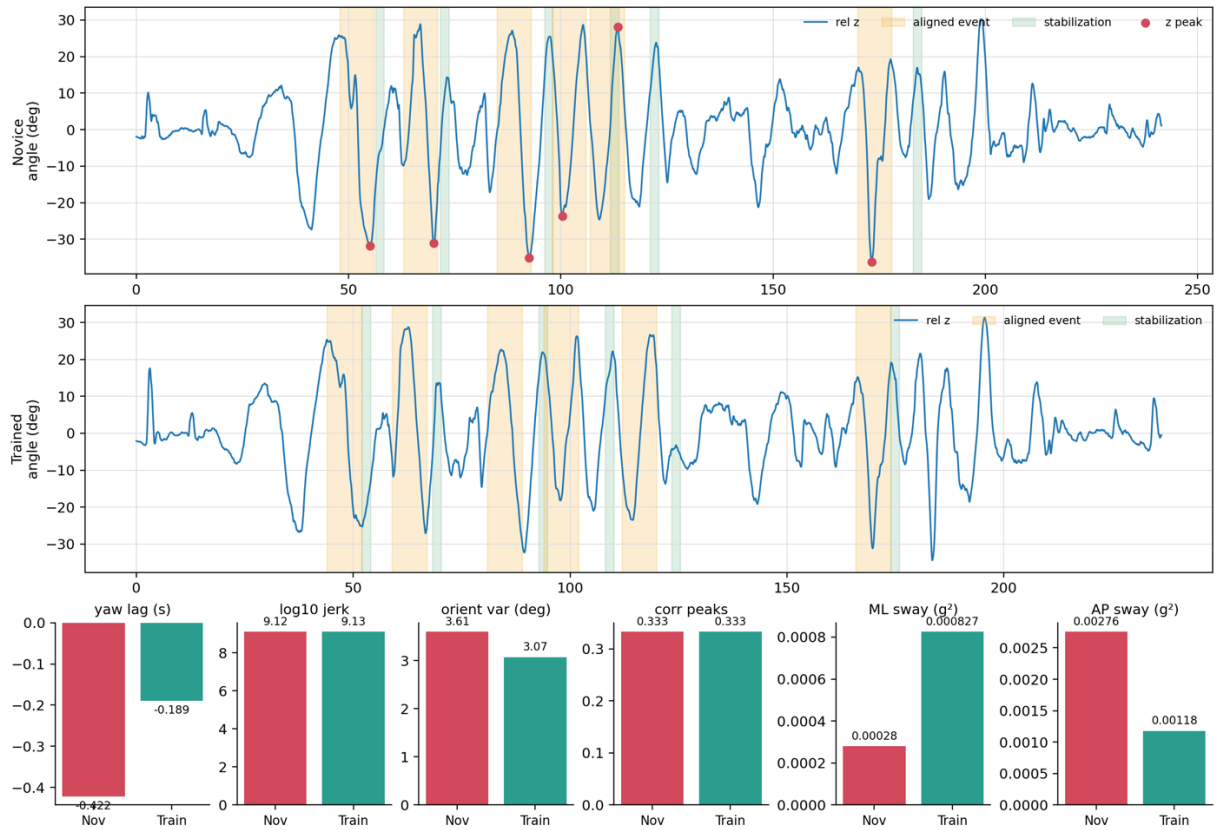
AI Usage Declaration:

A locally hosted instance of Qwen3.6 27B was used for inline code completion, code commenting assistance, debugging and guidance on correctly structuring variables for implementation within SciPy library functions.

AI agents were used to parse the IMU csv files into the analysis script and to adapt functions from my previous projects into the current tai chi analysis (madgwick sensor fusion, high and low pass filtering, and compute kinematics)

Additionally AI agents were used to quickly generate several additional visualization scripts (`plot_frame.py`, `plot_IMU.py`, and `plot_kinematics.py`). These scripts are non-critical supporting tools for data visualization and do not contribute to the core analysis.

Trunk Rotation Balance Traceability: Novice vs Trained Tai Chi



Monopodal Stance Balance Traceability: Novice vs Trained Tai Chi

